

Antonio H. Acuaviva

DOCTORAL RESEARCHER IN PURE MATHEMATICS · LANCASTER UNIVERSITY

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Education

Lancaster University

PHD IN PURE MATHEMATICS

Lancaster, UK

October 2023 – Present

- Supervisor: Professor Niels J. Laustsen.
- Research in functional analysis, with emphasis on Banach space theory and operator theory: complemented subspaces and primariness, $C(K)$ -spaces, operator ideals and Calkin algebras, and nonlinear Banach space theory.
- Awarded the **Pure Mathematics Prize** for exceptional performance during the first and third years of the PhD programme.
- EPSRC Doctoral Training Partnership Studentship (2023–present).

University of Cambridge

MASTER OF ADVANCED STUDY IN MATHEMATICS (PART III)

Cambridge, UK

October 2022 – June 2023

- Result: **Distinction**.
- Received the **1912 Senior Scholarship** and **Fitzwilliam College Prize** for academic performance.
- Supported by the **la Caixa Foundation Postgraduate Fellowship Abroad**.

Universidad Complutense de Madrid (UCM)

DOUBLE DEGREE IN MATHEMATICS AND PHYSICS

Madrid, Spain

September 2017 – June 2022

- Overall averages: **9.84/10 in Mathematics** (ranked **1st of 171 graduates**) and **9.66/10 in Physics** (ranked **1st of 247 graduates**).
- Recipient of the **Bachelor's Degree Extraordinary Award** in both Mathematics and Physics.
- Awarded the **Arturo Duperier Award** for the best academic record in Physics at UCM.
- Received **Community of Madrid Excellence Grants** in four consecutive years.
- Physics thesis: *Resilient Source Seeking with Robot Swarms*. Related journal article: *Fully distributed and resilient source seeking for robot swarms*.

University of California, Berkeley

NON-DEGREE EXCHANGE PROGRAMME (UCEAP)

Berkeley, USA

August 2019 – June 2020

- Selected for a year-long study program; recipient of the **UCM–University of California Scholarship**, awarded to the **top seven students** at Universidad Complutense de Madrid.

Publications

PUBLISHED JOURNAL ARTICLES IN ANALYSIS

- [1] **A. Acuaviva**. *The class of Banach lattices is not primary*. *Forum of Mathematics, Sigma*, 14:e41, 1–18 (2026).
- [2] **A. Acuaviva** and A. B. Nasser. *Quantifying (non-)weak compactness of operators on AL - and $C(K)$ -spaces*. *Proceedings of the Edinburgh Mathematical Society*, 1–56 (2026).
- [3] **A. Acuaviva**. *Complemented subspaces of a $C(K)$ -space constructed by Candido*. *Journal of Mathematical Analysis and Applications*, 556(1):130115 (2026) (11 pages).
- [4] **A. Acuaviva**. *Operators on injective tensor products of separable Banach spaces and spaces with few operators*. *Journal of Functional Analysis*, 290(1):111203 (2026) (17 pages).
- [5] **A. Acuaviva**. *Factorizations and minimality of the Calkin algebra norm for $C(K)$ -spaces*. *Journal of the London Mathematical Society*, 112(4):e70321 (2025) (20 pages).
- [6] **A. Acuaviva** and D. Seco. *Equidistribution of zeros of some polynomials related to cyclic functions*. *Analysis and Mathematical Physics*, 12(2):53 (2022) (24 pages).

OTHER PUBLISHED OR ACCEPTED JOURNAL ARTICLES

- [7] J. Bautista*, **A. Acuaviva***, J. Hinojosa, W. Yao, J. Jimenez, and H. Garcia de Marina. *Fully distributed and resilient source seeking for robot swarms*. *IEEE Transactions on Automatic Control*, Early Access, pp. 1–16 (2026).
*Equal contribution.
- [8] **A. Acuaviva**, A. Chan, S. Eldridge, C. Godsil, M. How-Chun-Lun, C. Tamon, E. Wright, and X. Zhang. *State transfer in chiral quantum walks*. *Philosophical Transactions of the Royal Society A: Mathematical, Physical and Engineering Sciences*, 383(2306):20240420 (2025) (19 pages).

PREPRINTS IN BANACH SPACE THEORY

- [9] **A. Acuaviva** and T. Kania. *Banach's isometric conjecture over the complex field*. arXiv:2608.18257 (2026) (17 pages).
- [10] **A. Acuaviva** and P. Acuaviva. *The uniform primary factorisation property for $C(K, E)$* . Preprint (2026) (16 pages).
- [11] **A. Acuaviva** and P. Acuaviva. *Mathematical discovery in the wild: AI-guided proofs in Banach space theory*. arXiv:2607.17388 (2026) (141 pages).
- [12] **A. Acuaviva** and P. Acuaviva. *A unital Banach algebra which is not a Calkin algebra*. arXiv:2607.15075 (2026) (21 pages).
- [13] **A. Acuaviva**. *A separable Banach space with a Schauder basis which is not a Lipschitz retract of its bidual*. arXiv:2607.12935 (2026) (21 pages).
- [14] **A. Acuaviva**, B. Horváth, and T. Kania. *Pure infiniteness and primary factorisation*. arXiv:2607.01467 (2026) (19 pages).
- [15] **A. Acuaviva**. *Preservation of primariness under ℓ_1^- , c_0^- , and ℓ_∞ -sums of Banach spaces*. arXiv:2606.26417 (2026) (39 pages).
- [16] **A. Acuaviva**. *Primariness of the spaces $\ell_p(C(K))$ for $1 \leq p \leq \infty$* . arXiv:2605.29854 (2026) (35 pages).
- [17] **A. Acuaviva** and T. Kania. *Primariness and the primary factorisation property*. arXiv:2605.21711 (2026) (74 pages).
- [18] **A. Acuaviva**. *$C(K)$ -spaces with few operators relative to posets*. arXiv:2511.22339 (2025) (26 pages).

OTHER PREPRINTS AND MANUSCRIPTS

- [19] **A. Acuaviva**, A. Acuaviva, and P. Acuaviva. *CNOT-distance is NP-complete under all-to-all connectivity*. arXiv:2608.03825 (2026) (14 pages).
- [20] **A. Acuaviva**, J. Esson, E. Kastis, and B. Schulze. *Curved RUM spectra of periodic frameworks*. Submitted manuscript (2026).

CONFERENCE PROCEEDINGS

- [21] **A. Acuaviva**, J. Bautista, W. Yao, J. Jimenez, and H. Garcia de Marina. *Resilient source seeking with robot swarms*. In *2024 IEEE 63rd Conference on Decision and Control (CDC)*, pp. 57–63 (2024).

Selected Research Experience

Fields Institute for Research in Mathematical Sciences

Toronto, Canada

UNDERGRADUATE RESEARCHER, FIELDS UNDERGRADUATE SUMMER RESEARCH PROGRAM

June 2021 – August 2021

- Investigated state transfer in quantum walks with Ada Chan, Christino Tamon, and Xiaohong Zhang; the project contributed to the later journal article *State Transfer in Chiral Quantum Walks*.
- Presented *Quantum Walks on Graphs: Perfect State Transfer and Periodicity* at the Canadian Undergraduate Mathematics Conference (2021).

Institute of Mathematical Sciences (ICMAT)

Madrid, Spain

INTRODUCTION TO RESEARCH GRANT RECIPIENT

June – September 2020 & 2021

- Received the competitive Severo Ochoa Introduction to Research Grant in two consecutive years and undertook projects in mathematical analysis.

Employment

Lancaster University

Lancaster, UK

GRADUATE TEACHING ASSISTANT (GTA)

October 2023 – December 2025

- Taught on Numbers and Relations (MATH111), Discrete Mathematics (MATH112), Real Analysis (MATH210), and Complex Analysis (MATH215), including marking, tutoring, and problem classes.

Honours & Awards

- 2024, 2026 **Pure Mathematics Prize**, Lancaster University
- 2025 **Spring Into Quant Finance Programme**, G-Research
- 2023 **EPSRC Doctoral Training Partnership Studentship**, Lancaster University
- 2023 **1912 Senior Scholarship**, Fitzwilliam College, University of Cambridge
- 2023 **Fitzwilliam College Prize**, Fitzwilliam College, University of Cambridge
- 2022 **Postgraduate Fellowship Abroad**, *la Caixa* Foundation
- 2022 **Bachelor's Degree Extraordinary Award**, Universidad Complutense de Madrid (Ranked 1st in class)
- 2022 **Department Research Grant**, Dept. of Mathematical Analysis, UCM
- 2022 **Blas Cabrera Summer Scholarship**, Universidad Internacional Menéndez Pelayo and Spanish Ministry of Science and Innovation
- 2022 **Arturo Duperier Award**, Pedro Bernardo City Council (best academic record in Physics at UCM)
- 2018–2021 **Excellence Grants**, Community of Madrid (awarded in four consecutive years)
- 2020 **Becas Atrévete 2020**, McKinsey & Company
- 2019 **UCM–University of California Scholarship**, Universidad Complutense de Madrid (top seven students selected)
- 2019 **Santander Progress Scholarship**, Santander Bank
- 2015–2017 **Goñi and Rey Foundation Fellowship**, Full academic funding

Languages

Languages English (Full Professional Proficiency), Spanish (Native), German (basic)